

Jay L. Karp

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PROFESSIONAL EXPERIENCE

Twitch (Amazon Web Services Interactive Video Service)

Jun 2022-Present

Senior Software Engineer

- Spearheaded and launched IVS Token Exchange by navigating complex dependencies across 6 microservices. Led application security, engineering teams and product teams to enhance customer security and UX, delivering the final product a full month ahead of the projected roadmap.
- Championed operational excellence, reducing on-call incident volume by 80% through a total overhaul of system observability. Simplified legacy code structures to eliminate technical debt and implemented standardized structured logging and metrics, enabling team to identify root causes in minutes rather than hours.
- Directed scalability initiatives for IVS RTX, increasing capacity from 10K to 500K concurrent viewers (CCV). Conducted investigations into microservice dependencies to harden systems for an eventual 1M CCV target.

Software Engineer II

- Architected and built a 30,000+ line notification and REST service from the ground up that processes thousands of TPS, migrating from DynamoDB streams and Lambda to ECS Fargate and SQS to reduce latency by 78%, while engineering all CloudFormation infrastructure, VPC networking, and CI/CD pipelines in multiple stages.
- Eliminated critical performance bottlenecks by developing a multi region edge authentication service in Go. Integrated Redis and DynamoDB to slash latency by 90%+ and implemented an SQS driven fanout architecture to scale the platform from dozens to 10k+ concurrent connections.

Software Engineer I

- Engineered public facing APIs integrated with the AWS ecosystem (SDKs). Implemented robust Smithy schemas within API Gateway and Go based services on ECS to deliver highly requested features for enterprise customers.
- Automated security validation for the entire IVS API ecosystem using Python and internal AWS security frameworks. Eliminated manual testing overhead, reducing the security audit workload for engineering teams by 10s of hours per project.

PRIOR EXPERIENCE

Apple, Software Engineering Intern

May 2021-Sep 2021

- Performed user centered design loop to build production quality tooling for 40 person customer feedback engineering team, collecting over 1 million pieces of feedback per year.
- Developed Ruby on Rails GraphQL API endpoints and built SwiftUI, AppKit and Apollo frontend application.
- Summarized and presented all work completed during internship to management in Keynote

Cheddar Inc., Full Stack and DevOps Intern

Jun 2019-Sep 2019

- Built full stack web application features for internet-based news outlet with 1 million+ monthly viewers.
- Used Ruby on Rails, GraphQL, and React to enhance web apis and serve refined content to frontend application.
- Wrote Golang API to interact with Prometheus pod inside of Kubernetes cluster.

TECHNICAL SKILLS

Languages: Golang, Typescript, Python, C, Haskell

AWS Services: DynamoDB, EC2, ECS, S3, IAM, VPC, DynamoDB, SNS/SQS, Kinesis, Lambda, API Gateway, Cloudwatch, Cloudformation

Architecture: Microservices, Event-Driven, Serverless, Distributed Systems, REST, gRPC, Websocket, GraphQL

DevOps: Docker, CI/CD pipelines, AWS CDK, Git, X-Ray, Terraform

Frontend: React, HTML, CSS, Swift & SwiftUI

EDUCATION

Columbia University

Masters in Computer Science: Columbia Fu School of Engineering

May 2022

BA in Computer Science: Columbia College

Apr 2021

Dean's List